

Multi-Asset Next-Day Return Prediction — Bitcoin, Gold, Silver

Final Year Project · Executive Summary for the Supervisor · Alyan Shahid
(teamlocalhost), University of Lahore

This project is a machine-learning system that forecasts the **next trading day's return** of three assets — Bitcoin (BTC-USD), Gold (GC=F futures) and Silver (SI=F futures) — and presents the forecast, together with its measured reliability, in a web dashboard and a REST API. The forecast target is the next-day log return $y_t = \ln(P_{t+1}/P_t)$; the price shown to the user is recovered as $P_t \cdot \exp(\hat{y})$. Daily data from Yahoo Finance (2018-01-01 to 2026-09-12; Bitcoin 3,147 usable days, Gold/Silver 2,156 exchange days) is enriched with macro series (US Dollar Index, WTI crude, 10-year Treasury yield, S&P 500, VIX) and the crypto Fear & Greed index.

Methodology. A frozen chronological split (train $\leq$ 2025-09-10, validation $\leq$ 2026-02-17, test = 2026-02-18 $\rightarrow$ 2026-09-12) with no shuffling; a scaler fitted on training rows only; 28–29 stationary, backward-looking features; hyper-parameters chosen by 4-fold expanding-window walk-forward validation inside train+val; and a single evaluation of the untouched test set. Ten model families are compared on identical days: a random-walk baseline, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM and a stacked ensemble. The **served forecast is the equal-weight combination** of the six trained models (no fitted weights, so nothing is selected on the test set), evaluated on the walk-forward folds and on the test set exactly like every single model. Every feature obeys a *close-time rule* — it may only use what is known when the asset's daily close is fixed (same-day macro data for Bitcoin, previous-day data for the metals, whose Yahoo close is the 13:30 ET COMEX settlement). Fourteen specific data-leakage checks are enforced by a 41-case test suite; only complete daily bars are ever used.

Result. On the untouched test set all three assets sit at the random-walk floor: the served combined forecast's return-RMSE is within ± 0.3 % of the naive forecast (Diebold–Mariano $p = 0.49 / 0.97 / 0.46$), directional accuracy 46.9 % / 50.3 % / 53.8 % (binomial $p = 0.83 / 0.50 / 0.20$) and R^2 in return space ≈ 0 ; the best single models by validation (CatBoost for Bitcoin and Gold, a GRU for Silver) are 0.0–0.6 % *above* the naive forecast. **No model — single or combined — beats the random walk at a one-day horizon for Bitcoin, Gold or Silver.** The result is consistent with weak-form market efficiency; the application reports the measured error next to every forecast. An intermediate version of the system reported 62 % directional accuracy for Silver; the final audit traced it to a close-time leak (same-day macro features against a 13:30 ET futures settlement), fixed it and re-ran the pipeline — the finding is reported in Section 9.

Improvement over the initial version. An audit of the original codebase found a price-reconstruction bug that corrupted every dollar metric, 30 % fabricated weekend rows for the metals, hyper-parameters selected on the test set, early stopping on training data, non-stationary price-level features, a served ensemble that predicted a constant, a broken inference path and misleading R^2 values (≈ 0.95 on price levels, which a random walk also achieves). All were fixed. A second, measured improvement phase then added $2.9\times$ more training data (2018 $\rightarrow$), volatility/momentum features, re-tuning, and logged design experiments; a final audit found and removed the close-time leak. On identical unseen days the best single models moved by -0.6 % (Gold), -0.4 % (Silver) and $+0.1$ % (Bitcoin) in return-RMSE — inside noise: more data made the models equal to the random walk rather than slightly worse, not better than it.

Status. Working Streamlit dashboard (one-click combined forecast with uncertainty band and every model's own prediction, a *Predict-a-Day* demonstration on the unseen test period that reveals the actual price and the error, the full test-set history, validation/test tables, regime analysis and experiment tables) and FastAPI endpoints; reproducible pipeline (make pipeline); documentation for methodology, features, results, limitations, demonstration and viva.

Key facts

Item	Final implementation
Assets / horizon	Bitcoin, Gold (GC=F), Silver (SI=F); next trading day
Target	log return $\ln(P_{t+1} / P_t)$; price = $P_t \cdot \exp(\hat{y})$
Data	Yahoo Finance daily OHLCV 2018-01-01 → 2026-09-12 (BTC 3,147 / metals 2,156 usable days) + DXY, WTI, 10-y yield, S&P; 500, VIX, Fear & Greed
Features	29 (BTC) / 28 (Gold) / 29 (Silver) stationary, backward-looking features incl. HAR/EWMA volatility and momentum; close-time rule (previous-day macro for the metals); raw price levels never used as inputs
Split	train $\leq$ 2025-09-10 · validation $\leq$ 2026-02-17 · unseen test 2026-02-18 → 2026-09-12 (207 / 143 / 143 days), chronological, evaluated once
Validation	4-fold expanding-window walk-forward on train+val for tuning and model selection
Models	Naive random walk, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM, stacked ensemble
Served forecast	Equal-weight combination of the six trained models (evaluated on the folds and on the test set like every model); best single models by CV: CatBoost for Bitcoin and Gold, GRU for Silver (all within 0.35 % of the random walk on validation)
Metrics	MAE/RMSE/MAPE (\$), RMSE/MAE/R ² (return), directional accuracy + binomial p, Diebold–Mariano vs random walk, strategy backtest
Stack	Python 3.9, pandas, scikit-learn, LightGBM, CatBoost, TensorFlow/Keras, statsmodels, Streamlit, FastAPI, Docker, pytest (41 cases), GitHub Actions
Demo	One-click combined forecast with uncertainty band; Predict-a-Day on the unseen test period (predicted vs actual vs error vs HIT/FAIL); test-set history

Headline test-set results

Asset	Model	MAE (\$)	RMSE (\$)	MAPE %	R ² (return)	Dir. acc. % (p)	DM p vs naive
Bitcoin	Naive (random walk)	1,066.42	1,451.73	1.52	-0.001	—	—
Bitcoin	Combined (mean of 6) (served)	1,063.94	1,447.37	1.52	+0.006	46.9 (0.83)	0.49
Bitcoin	CatBoost (best single)	1,071.07	1,453.76	1.53	-0.004	46.9 (0.83)	0.74
Gold	Naive (random walk)	58.26	75.76	1.29	-0.002	—	—
Gold	Combined (mean of 6) (served)	58.20	75.73	1.29	-0.001	50.3 (0.50)	0.97
Gold	CatBoost (best single)	58.43	76.22	1.30	-0.015	51.7 (0.37)	0.71
Silver	Naive (random walk)	1.79	2.37	2.49	-0.001	—	—
Silver	Combined (mean of 6) (served)	1.78	2.37	2.47	+0.005	53.8 (0.20)	0.46
Silver	GRU (best single)	1.79	2.37	2.48	-0.001	55.2 (0.12)	0.99

Unseen test period 2026-02-18 → 2026-09-12. Full 11-row tables: [results/FINAL_RESULTS.md](#) and the main report.

Reading the numbers

R² is reported on returns, where 0 means "as good as predicting the mean". The original project reported R² ≈ 0.95 on price levels — a value the random walk also achieves, which is why it was misleading. The served combined forecast's errors do not differ significantly from the random walk's for any asset (DM p = 0.49 Bitcoin, 0.97 Gold, 0.46 Silver; direction 47 / 50 / 54 %, none significant). An intermediate version showed 62 % for Silver; the final audit traced it to same-day macro features being used against a 13:30 ET futures settlement close, fixed the alignment and re-ran everything. On identical unseen days the improvement phase moved return-RMSE by -0.6 % (Gold), -0.4 % (Silver) and +0.1 % (Bitcoin) — inside noise; 2.9× more data made the models equal to the random walk, not better than it.

Current status: implementation complete; pipeline reproducible; dashboard (with the unseen-day demonstration) and API verified running for all three assets; documentation, demo guide and viva notes prepared. **Recommended framing:** a leakage-audited forecasting benchmark and decision-support prototype whose predictions can be verified day by day, and which found and removed a subtle leak in its own intermediate results — not a market-beating predictor.